

Quadratic formula



- 1 The standard form of a quadratic equation is $ax^2 + bx + c = 0$. Identify the values of a , b and c in the following quadratic equations:

a $8x^2 - 3x = 0$

b $8x^2 + 3 = 0$

c $5x^2 + 6x = 10$

d $3x^2 - 8x + 2 = 9x - 7$

- 2 Solve the following equations by using the quadratic formula:

a $x^2 + 11x + 28 = 0$

b $4x^2 + 6x + 2 = 0$

c $4x^2 - 7x - 15 = 0$

- 3 To solve $x^2 - x = 2$, Emma substitutes $a = 1$ and $b = -1$ into the quadratic formula. What value of c should she substitute into the quadratic formula?

- 4 Consider the quadratic equation $9y^2 = 8y$. Suppose $b = 8$ is used in the quadratic formula:

- a** State the value of a that should be used. **b** State the value of c that should be used.

- 5 Solve the following equations below using quadratic formula, leaving your answer in radical form.

a $x^2 - 5x - 2 = 0$

b $4x^2 - x - 10 = 0$

c $-2x^2 - 15x - 4 = 0$

- 6 Solve the following equations below using quadratic formula, leaving your answers as decimals to the nearest thousandth.

a $x^2 + 7x - 3 = 0$

b $5x^2 + 9x + 2 = 0$

- 7 Solve the equation $1.8x^2 + 5.2x - 2.3 = 0$ using quadratic formula, leaving your answers as decimals to the nearest tenth.

- 8 Consider the equation $2x^2 = 14$.

- a** Use the quadratic formula to find the solutions of the equation. Give your answer as an exact value.
b Give the solutions as decimals rounded to the nearest tenth.

9 Solve the following equation by using the quadratic formula.



a $x(x - 4) = 5$

b $x(x + 9) = -20$

Discriminant

10 For a particular quadratic equation, $b^2 - 4ac = 0$. What does this tell us about the solutions of the quadratic?

11 The solutions of a quadratic equation are 9 and -9 . What does this tell us about the value of $b^2 - 4ac$?

12 For each of the following equations:

i Find the value of the discriminant.

ii State the number of real solutions the equation has.

a $x^2 + 6x + 9 = 0$

b $4x^2 - 6x + 7 = 0$

c $2x^2 - 2x = x - 1$

13 State whether the following equations have any real solutions:

a $-x^2 - 8x - 16 = 0$

b $-2x^2 + 5x - 6 = 0$

c $-x^2 + 6x + 1 = 0$

14 Consider the equation $4x^2 - 4x + 1 = 0$.

a Find the discriminant.

b Now, describe the nature of the roots.

15 Consider the equation $x^2 - 8x - 48 = 0$.

a Find the discriminant.

b Now, describe the nature of the roots.

c Find the solutions of the equation.



Quadratics with complex solutions

- 16** Below is the result after using the quadratic formula to solve a quadratic equation:

$$x = \frac{10 \pm \sqrt{-1}}{8}$$

How many real solutions does the quadratic equation have?

- 17** Below is the result after using the quadratic formula to solve a quadratic equation:

$$x = \frac{-3 \pm \sqrt{9 - 4 \times 56}}{16}$$

How many real solutions does the quadratic equation have?

- 18** Solve the following, leaving your answers in the form $a \pm bi$:

a $x^2 + 2x + 5 = 0$

b $25 = x(6 - x)$

c $x^2 + 10x + 28 = 0$

d $x^2 - 4x + 29 = 0$

e $x^2 - 10x + 27 = 0$

f $4x^2 + 20x + 33 = 0$

g $-4x^2 + 4x - 13 = 0$

h $12 - 22x + 5x^2 = 4 - 4x^2 - 10x$

i $\frac{-10x - 40}{x} = x$

j $\frac{x^2}{2} = 2x - 18$

k $9 + \frac{115}{x^2} = \frac{-24}{x}$

- 19** Find the zeros of the function $8x^2 + 7x + 3 = 0$ algebraically, leaving your answer in the form $a \pm bi$.

- 20** Consider the quadratic equation $ax^2 + 2x + c = 0$. State whether the following pairs of values for a and c will give non-real solutions:

a $a = -5$ and $c = -6$

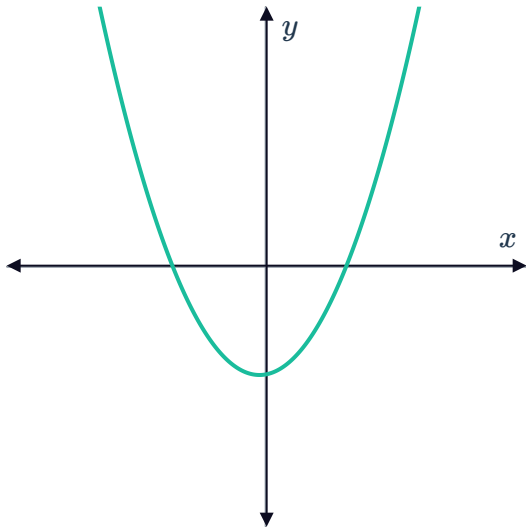
b $a = -5$ and $c = 6$

c $a = 5$ and $c = -6$

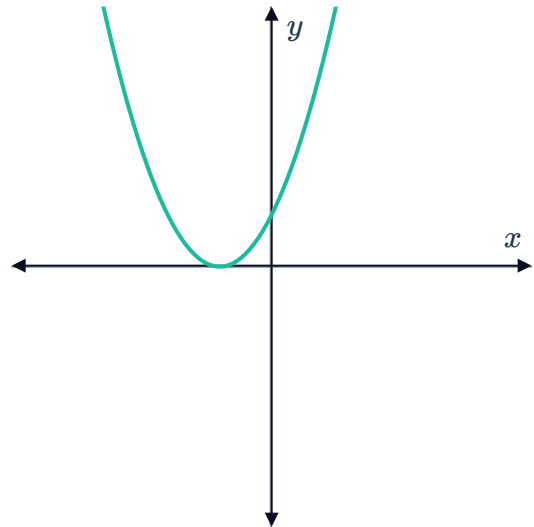
d $a = 5$ and $c = 6$

21 How many non-real solutions do the equation  the following quadratic curves have?

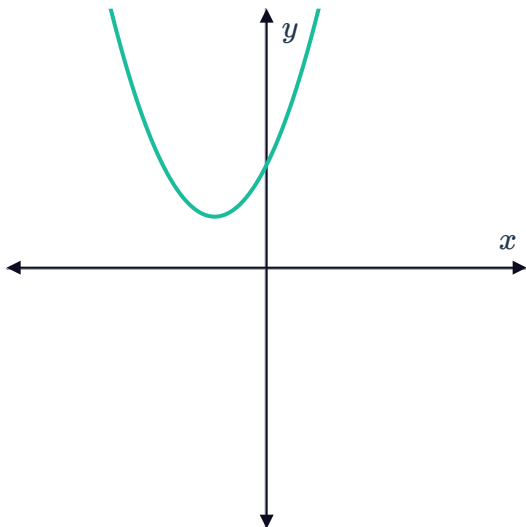
a



b



c



22 Consider the equation in x : $mx^2 + 2x - 1 = 0$. Given that it has two solutions, state the possible values of m .

23 Find the values of n for which $x^2 - 8nx + 1296 = 0$ has one solution.

24 Find the value of k for which $16x^2 + 8x + k = 0$ has equal roots.

25 Consider the equation $2x^2 + 8x + k = 0$.



- a** Find the discriminant in terms of k .
- b** Find the value of k when the equation has:
 - i** Equal solutions
 - ii** Real solutions
 - iii** No real solutions.
 - iv** Real and distinct solutions.

26 Consider the equation $(k + 4)x^2 + 10x + 3 = 0$.

- a** Find the discriminant in terms of k .
- b** Find the value of k when the equation has:
 - i** Equal solutions
 - ii** Real solutions
 - iii** No real solutions
 - iv** Real and distinct solutions

Discriminant and parabolas

27 To find the x -intercepts of a particular parabola, $ax^2 + bx + c = 0$, Katrina used the quadratic formula and found that $b^2 - 4ac = -5$. How many x -intercepts does the parabola have?

28 The graph of $y = m - 9x - 3x^2$ has no x -intercepts.

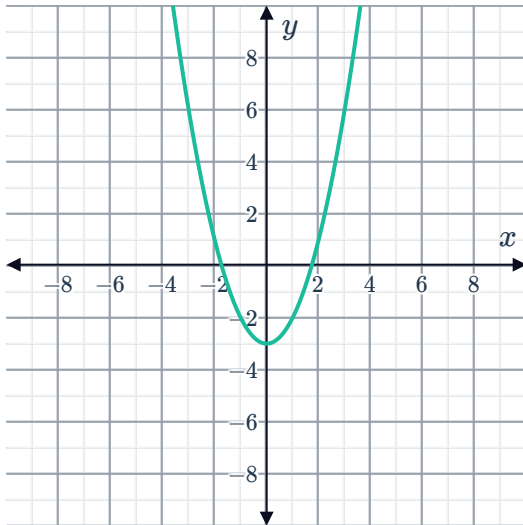
- a** Solve for the possible values of m .
- b** What is the largest possible integer value of m ?

29 Solve for the value(s) of k for which $y = 4x^2 - 4x + k - 15$ just touches the x -axis.

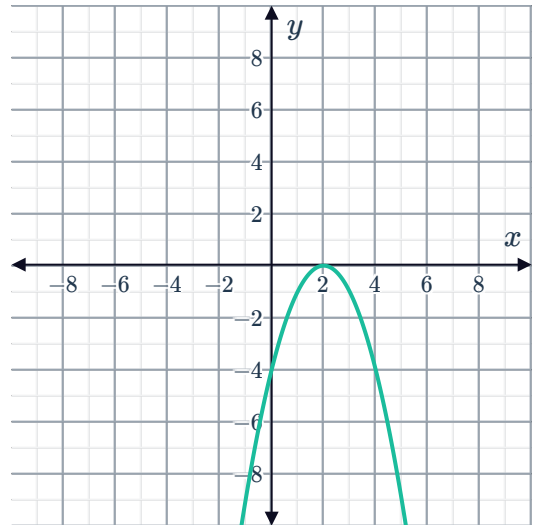
30 For each of the following graphs of a quadratic function $f(x) = ax^2 + bx + c$, with discriminant $b^2 - 4ac$:

- State whether $a > 0$ or $a < 0$.
- State whether $b^2 - 4ac > 0$, $b^2 - 4ac < 0$ or $b^2 - 4ac = 0$.

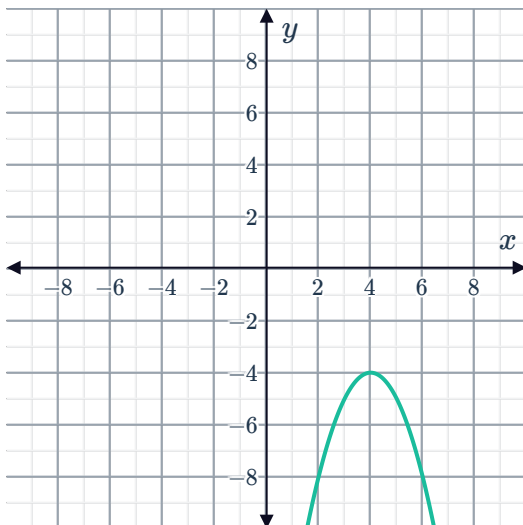
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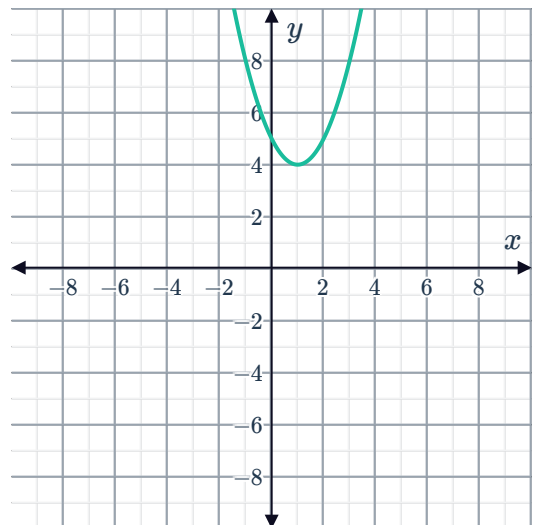
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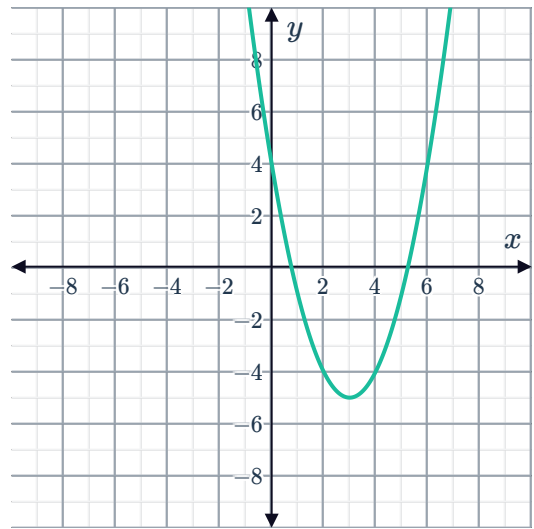
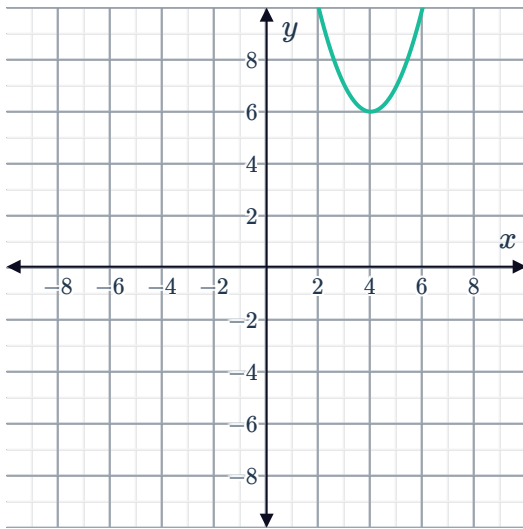
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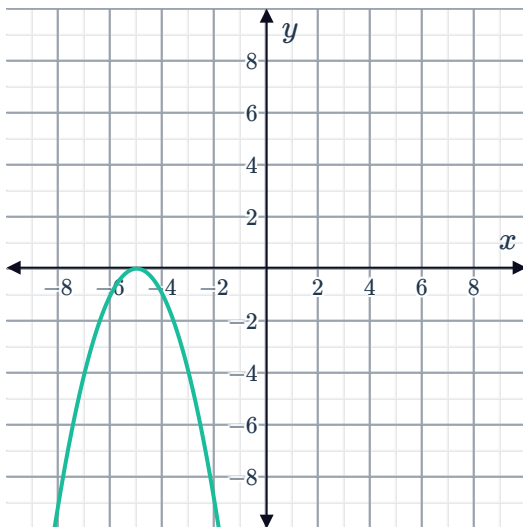
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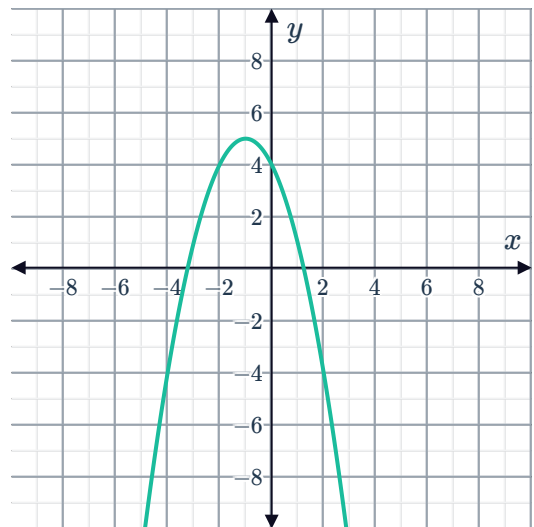
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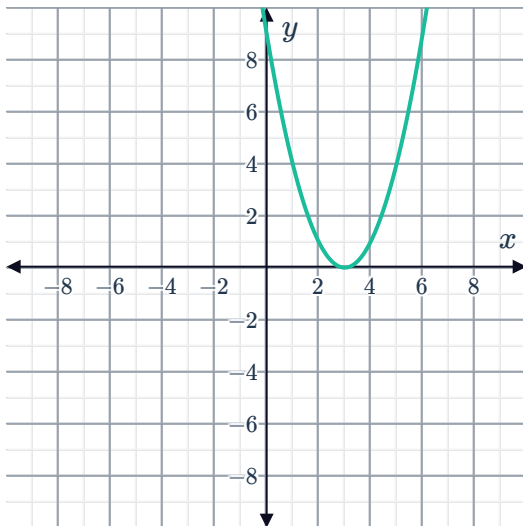
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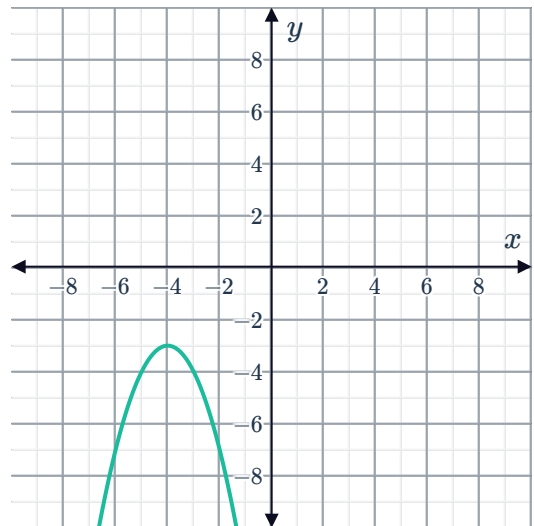
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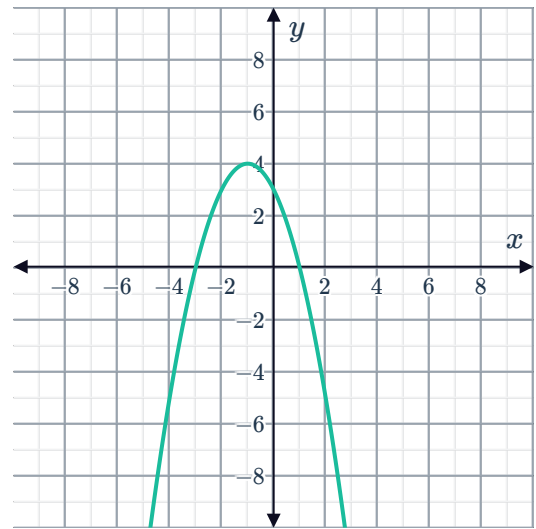
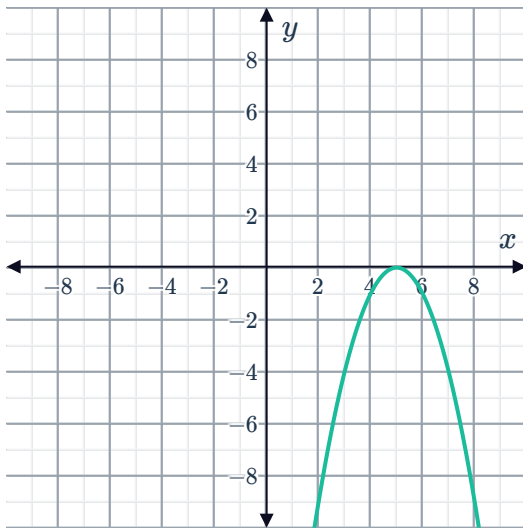
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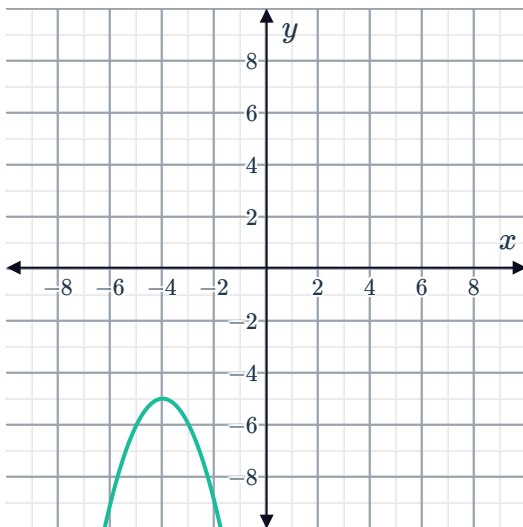
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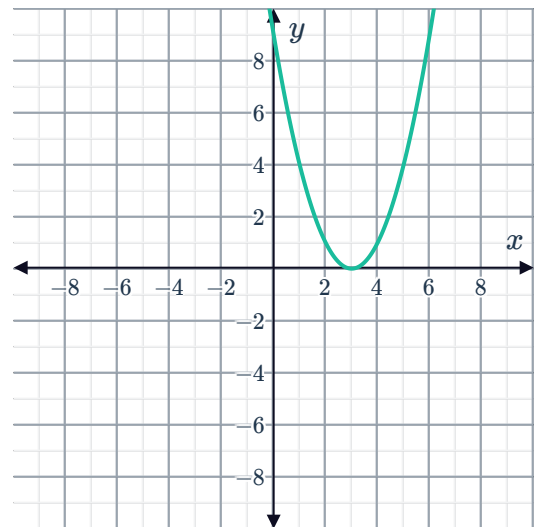
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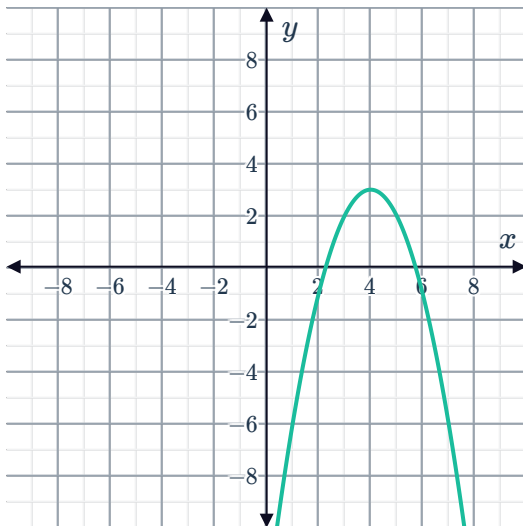
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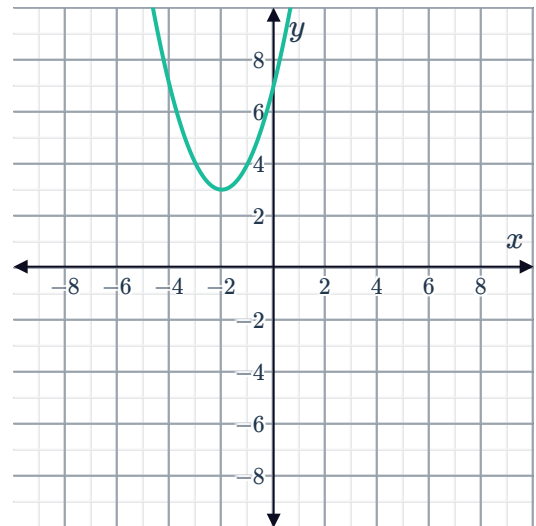
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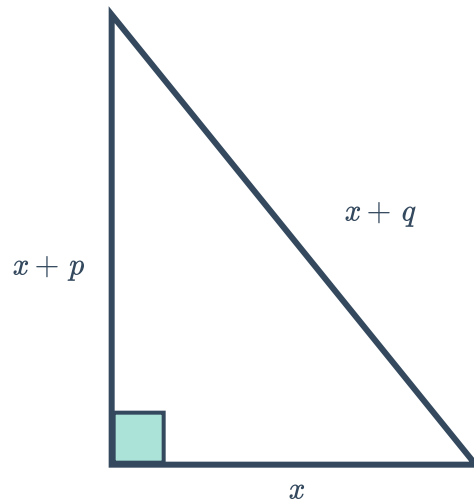




Applications

31 Consider a right-angled triangle with side lengths x units, $x + p$ units and $x + q$ units, ordered from shortest to longest. No two sides of this triangle have the same length.

- a** Complete the statement:
 p and q have lengths such that
 $0 < p < q$.
- b** Write a quadratic equation that describes the relationship between the sides of the triangle in terms of x .
- c** Find the discriminant of this quadratic equation.
- d** How many real solutions are there to the quadratic equation?
- e** Find the value of x when $q = 2p$. Give your answer in terms of p .



32 A cannon is fired from the top of a cliff 26 meters above sea level. The height of the cannon-ball above sea level, in meters, after t seconds have passed is given by the formula:

$$h = -5t^2 + 56t + 26$$

- a** If we want to find out when the cannon-ball will reach a height of 200 meters, what equation do we need to solve?
- b** Solve the equation found in part (a) for t .
- c** Does the cannon-ball reach a height of 200 meters? Explain your answer.

33 An object is launched from a height of 90 feet with an initial velocity of 131 feet per second, and after x seconds its height (in feet) is given by $h = -16x^2 + 131x + 90$. Solve for the number of seconds x after which the object is 20 feet above the ground. Round your answer to the nearest tenth of a second.